

Why is this the test statistics for testing hypotheses for proportions?

For a binomial random variable $E(X) = np$ and $V(X) = np(1 - p)$.

Test statistic for proportions:

$$z = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}} \quad (1)$$

where $\hat{p} = \frac{X}{n}$ (sample proportion),

p is the population proportion, and n is the sample size.

The formula above for the test statistics can be derived from the $z = \frac{X - \mu}{\sigma}$ formula. Since $\mu = np$ and $\sigma = \sqrt{np(1 - p)}$, we can write the following:

$$z = \frac{X - np}{\sqrt{np(1 - p)}} = \frac{X/n - np/n}{\sqrt{np(1 - p)/n}} = \frac{\hat{p} - p}{\sqrt{np(1 - p)/n^2}} = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}}.$$